

Two fair pure densities with polynomial-denominator cell means

Complete proof; three independent reviews passed

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Here and below “polynomial” means bounded by one fixed power of $m + 1$. Constants and exponents are absolute; their optimization is not attempted. This note strengthens the rational two-density completion by controlling one common denominator for all cell masses and first moments.

Theorem 1. *For every $m \geq 2$ there are rational piecewise polynomial probability densities f_A, f_B on $[0, 1]$ such that A, B together with m independent uniforms are fully permutation-fair. There is a partition into polynomially many rational cells with these additional properties:*

1. *On each open cell exactly one of f_A, f_B is positive. Its conditional density, affinely rescaled to $[0, 1]$, lies between two absolute positive constants and has polynomial degree.*
2. *Every cell length and its positive mass are at least an inverse polynomial.*
3. *All endpoints, all cell masses, and all unnormalized first cell moments have one common polynomial rational denominator.*

No denominator bound is asserted for the density coefficients or higher cell moments.

Lemma 1 (Conditioning cumulative mass and first moment). *Let $J \geq 2$, $d > 2(J + 4) + 1$ with $d = O(J)$, and put*

$$H(t) = \int_0^t P_d(2u - 1)du, \quad R_0(t) = t - \eta H(t), \quad R_1(t) = \int_0^t u[1 - \eta P_d(2u - 1)]du,$$

where $0 < \eta \leq 1/2$. The family $\{\phi_j, R_0\phi_j, R_1\phi_j : 0 \leq j < J\}$, where ϕ_j are orthonormal shifted Legendre polynomials, has Gram matrix bounded below by $\eta^2(J + 1)^{-C}$ for an absolute C . The same holds on separated midpoint-like exact quadrature nodes of sufficient degree, and under sufficiently small polynomial perturbations.

Proof. Write $w = t(1 - t)$ and $\lambda = d(d + 1)$. The primitive identity

$$\int_0^t H(u)du = \frac{w'H - wH'}{\lambda - 2}$$

gives

$$R_1 = \frac{t^2}{2} - \eta \left(t - \frac{w'}{\lambda - 2} \right) H - \frac{\eta w}{\lambda - 2} H'.$$

Thus $p + qR_0 + vR_1$ is $\tilde{p} + H\tilde{q} + H'\tilde{v}$, with $\tilde{v} = -\eta wv/(\lambda - 2)$, and all coefficient degrees at most $J + 2$. The three-band lemma in `../rational_designs/asymptotic/ordered_prefix_legendre_tools.tex` recovers these coefficients with a polynomial norm loss. Multiplication by w on $\mathcal{P}_{J-1}$ has inverse norm at most $C(J + 1)^2$: delete endpoint intervals of length $c(J + 1)^{-2}$ and apply Nikolskii to retain half the squared norm. Recover v , then q , then p . The total inverse norm is at most $\eta^{-1}(J + 1)^C$. An exact quadrature for the squares reproduces this Gram matrix. Polynomial evaluation and derivative bounds preserve it under sufficiently small perturbations. $\square$

Proof of Theorem 1. Set $J = m+3$, choose d as in Lemma 1, and choose $\eta = (m+1)^{-A}$ with A a sufficiently large fixed integer to be specified by the positivity estimates below. Take a separated real equal-weight rule b_q^0 , $1 \leq q \leq K$, of degree at least $2(d+J+3)$, with $K = C(m+1)^4$ rounded to an integer and midpoint discrepancy $O(1/K)$. Such a rule, with endpoint and consecutive gaps at least c/K , is provided by the midpoint-grid Newton lemma in `../rational_designs/asymptotic/two_discrete_step_compatibility.tex`. The exact degree can be increased by an absolute factor without changing this bound on K .

Legendre orthogonality and Fubini give, for $0 \leq j \leq J$,

$$\frac{1}{K} \sum_q (b_q^0)^j = \frac{1}{j+1}, \quad \frac{1}{K} \sum_q R_0(b_q^0)(b_q^0)^j = \frac{1}{j+2}, \quad \frac{1}{K} \sum_q R_1(b_q^0)(b_q^0)^j = \frac{1}{2(j+3)}. \quad (1)$$

For the third identity the correction integral is $-(\eta/(j+1)) \int_0^1 u P_d(2u-1)(1-u^{j+1})du = 0$. In particular the two feature means are $1/2$ and $1/6$.

Choose one polynomial integer L , divisible by 6, with sufficiently large fixed exponent. Independently round the arrays $R_0(b_q^0)$ and $R_1(b_q^0)$ to $r_{0q}, r_{1q} \in L^{-1}\mathbb{Z}$, preserving their sums $K/2$ and $K/6$ exactly. Balanced rounding gives error at most $1/L$. For the remaining equations in (1), keep r_0, r_1 fixed and vary the b_q . Use antiderivatives of orthonormal polynomials as the nonconstant moment basis. The derivative matrix has rows given by $K^{-1}\phi_j(b_q)$, $K^{-1}r_{0q}\phi_j(b_q)$, and $K^{-1}r_{1q}\phi_j(b_q)$, $j < J$. Lemma 1 provides a right inverse of polynomial norm because η is an inverse polynomial. The rounding residual and Jacobian perturbation are at most a polynomial times $1/L$. Polynomial derivative bounds give a fixed-inverse contraction in a ball of radius $\text{poly}(m)/L$. Taking the exponent in L sufficiently large gives real separated nodes b_q satisfying

$$K^{-1} \sum_q b_q^j = \frac{1}{j+1}, \quad K^{-1} \sum_q r_{0q} b_q^j = \frac{1}{j+2}, \quad K^{-1} \sum_q r_{1q} b_q^j = \frac{1}{2(j+3)} \quad (0 \leq j \leq J), \quad (2)$$

with $\max_q |b_q - b_q^0| \leq \text{poly}(m)/L$. The empirical feature Gram bound survives. In particular

$$|r_{0q} - b_q| + |r_{1q} - b_q^2/2| \leq C\eta$$

after increasing L . The array r_0 is strictly increasing and interior, since its baseline has derivative $1 - \eta P_d \geq 1/2$.

Apply the mean-preserving rational box theorem in `../rational_designs/asymptotic/polynomial_mean_box_smoothing.tex` with the three fixed features $(1, r_0, r_1)$ and degree J . Its low-degree arithmetic targets are the fixed rational numbers $1/2, 1/3, 1/8$, so its hypotheses hold. Obtain rational boxes $J_q = [c_q - \epsilon, c_q + \epsilon]$, each of mass $1/K$, with positive quadratic conditional densities. Their endpoints and means c_q lie on one polynomial rational grid. All three families in (2) remain exact with box integrals in place of node evaluations. The exponent in the rational halfwidth ϵ can be chosen arbitrarily large in advance, and the center displacement can be made smaller than any prescribed inverse polynomial. The complementary gaps still have lengths at least $\Delta = c/K$. Let this smoothed density be $f_B = \beta$.

On the complementary gaps prescribe cumulative mass r_{0q} and cumulative first moment r_{1q} before box q . At the two outer endpoints prescribe $(0, 0)$ and $(1, 1/2)$. Let g be the unique affine density on each gap with the resulting mass and first moment, and zero on the boxes. All its coefficients are rational. The elementary rescaled two-moment system gives

$$\|g - 1\|_{L^\infty(G)} \leq C\Delta^{-2}(\eta + \epsilon + \max_q |c_q - b_q|),$$

so it lies between $1/2$ and $3/2$ when the exponents are large enough. The weighted identities (2) give precisely compatibility on $\mathcal{S}_1$ in `quantitative_smoothed_quotient.tex`. That note gives

$$\|1 - g\|_2 \leq C\Delta^{-2}(\eta + \epsilon + \max |c - b|) + \sqrt{2K\epsilon},$$

and its completion applies if this is at most $c(m+1)^{-7}\Delta^3$ and $\epsilon \leq c\Delta/(m+1)^3$. Choose the fixed exponent A first sufficiently large, then the box width exponent, then the rounding precisions sufficiently large. All choices remain polynomial, and these inequalities follow. We obtain a rational piecewise polynomial density f_A , between $1/4$ and $7/4$ on the gaps, with exactly the prescribed mass and first moment on every gap, and

$$\int f_A(a)F(a)da = \int_0^1 F(a)da \quad (F \in \mathcal{P}_m + \text{span}\{a^r S_s(a) : r+s \leq m\}).$$

Consequently all ordered polynomial moments of total degree at most m of (A, B) agree with those of two independent uniforms. For example

$$\mathbb{E}[A^r B^s \mathbf{1}_{A < B}] = \int f_A(a) a^r S_s(a) da = \frac{1}{r+1} \int b^{r+s+1} \beta(b) db = \frac{1}{(r+1)(r+s+2)}.$$

The other chamber follows from the marginal moments and independence. Integrating the m uniform variables in a fixed full order leaves a polynomial of total degree m on either chamber, proving full fairness.

Finally, every A -gap mass and first moment is a difference of entries of r_0, r_1 , hence has denominator dividing $2L$. Every B -box mass is $1/K$ and its first moment is c_q/K . The endpoints and means c_q have one polynomial common denominator by the smoothing theorem. Multiplying these finitely many *common* denominators, rather than individual cell denominators, yields the asserted single polynomial grid. The gap and box lengths are inverse polynomials. The density bounds imply the same for every positive cell mass and give the claimed conditional density bounds after rescaling. The piece degrees are at most m on the gaps and two on the boxes. $\square$